Complete Guide: Permanently Stop a Windows Process

Step-by-step instructions to identify, stop, and disable any process

📦 Example: msiexec.exe

⚡ Pre-requisite: Run as Administrator

All commands in this guide require elevated privileges. You must run Command Prompt as Administrator.

🔐 How to Open Admin Command Prompt

1. Press `Win` + `S` to open search
2. Type `cmd`
3. Right-click on **Command Prompt**
4. Select **"Run as administrator"**
5. Click **Yes** on the UAC prompt

💡 ALTERNATIVE METHODS

```
• Press Win + X → Select "Terminal (Admin)" or "Command Prompt (Admin)"
• Press Win + R → Type cmd → Press Ctrl + Shift + Enter
```

1 Identify the Process

First, confirm the process is running and get its details.

📌 **Example Process**

For this guide, we will stop the process `msiexec.exe` (Windows Installer) as our example.

> **Command Prompt (Admin)**

```
tasklist /FI "IMAGENAME eq msiexec.exe"
```

📤 **OUTPUT**

```
Image Name       PID     Session Name     Session#     Mem Usage
msiexec.exe      4532    Services         0            8,456 K
```

2 Find the Parent Process

Identify what spawned this process to understand its origin.

> Command Prompt (Admin)

```
wmic process where name="msiexec.exe" get ProcessId,ParentProcessId,C
```

📤 **OUTPUT**

```
CommandLine                                ProcessId    ParentProcessId
C:\Windows\system32\msiexec.exe /V         4532         672
```

Now find the parent process (PID 672):

> Command Prompt (Admin)

```
wmic process where ProcessId=672 get Name,CommandLine
```

📤 **OUTPUT**

```
CommandLine    Name
               services.exe
```

💡 **What this tells us**

The process is spawned by **services.exe** (Windows Services), meaning it's managed by a Windows Service.

3 Identify the Associated Service

Find which service owns this process.

> Command Prompt (Admin)

```
tasklist /SVC /FI "IMAGENAME eq msiexec.exe"
```

📤 **OUTPUT**

```
Image Name      PID      Services
msiexec.exe     4532     msiserver
```

🎯 **Service Identified**

The service name is **msiserver** (Windows Installer).

4 Check Service Status

Verify the current state of the service.

> Command Prompt (Admin)

```
sc query msiserver
```

📤 **OUTPUT**

```
SERVICE_NAME: msiserver
    TYPE               : 10  WIN32_OWN_PROCESS
    STATE              : 4  RUNNING
    WIN32_EXIT_CODE    : 0
    CHECKPOINT         : 0
    WAIT_HINT          : 0
```

5 Stop the Service

Stop the service to terminate the process gracefully.

> Command Prompt (Admin)

```
net stop msiserver
```

📤 **OUTPUT**

```
The Windows Installer service is stopping.
The Windows Installer service was stopped successfully.
```

6 Disable the Service

Prevent the service from auto-restarting on reboot.

> Command Prompt (Admin)

```
sc config msiserver start= disabled
```

📤 **OUTPUT**

```
[SC] ChangeServiceConfig SUCCESS
```

⚠️ **Important Note**

The space after `start=` is required. The syntax is `start= disabled` not `start=disabled`

7 Kill Remaining Process Instances

Force terminate any remaining process instances.

> Command Prompt (Admin)

```
taskkill /IM msiexec.exe /T /F
```

Flag	Meaning
/IM	Image name (process name)
/T	Kill entire process tree (parent + children)
/F	Force termination

8 Verify Process is Gone

Confirm the process has been completely terminated.

> Command Prompt (Admin)

```
tasklist /FI "IMAGENAME eq msiexec.exe"
```

📤 **EXPECTED OUTPUT**

```
INFO: No tasks are running which match the specified criteria.
```

9 Verify Service is Disabled

Confirm the service is stopped and disabled.

> Command Prompt (Admin)

```
sc query msiserver
```

📤 **EXPECTED OUTPUT**

```
SERVICE_NAME: msiserver
    STATE              : 1  STOPPED
```

Check startup type:

> Command Prompt (Admin)

```
sc qc msiserver
```

📤 **LOOK FOR**

```
START_TYPE         : 4   DISABLED
```

📋 Quick Reference - Complete Sequence

> Complete Command Sequence

```
:: ═══════════════════════════════════════════════════════════
::  PERMANENTLY STOP msiexec.exe - Complete Sequence
:: ═══════════════════════════════════════════════════════════

:: Step 1-3: IDENTIFY
tasklist /FI "IMAGENAME eq msiexec.exe"
wmic process where name="msiexec.exe" get ProcessId,ParentProcessId
tasklist /SVC /FI "IMAGENAME eq msiexec.exe"

:: Step 4: CHECK STATUS
sc query msiserver

:: Step 5: STOP SERVICE
net stop msiserver

:: Step 6: DISABLE SERVICE
sc config msiserver start= disabled

:: Step 7: KILL REMAINING INSTANCES
taskkill /IM msiexec.exe /T /F

:: Step 8-9: VERIFY
tasklist /FI "IMAGENAME eq msiexec.exe"
sc query msiserver
```

🔄 To Re-Enable Later

> **Restore Commands**

```
:: Enable the service
sc config msiserver start= demand

:: Start the service
net start msiserver
```

Start Type	Value	Meaning
auto	2	Starts automatically on boot
demand	3	Manual start only
disabled	4	Cannot start

📚 Common Self-Restarting Processes

Process	Service to Stop
`msiexec.exe`	msiserver
`svchost.exe`	Multiple (check with `tasklist /SVC`)
`WmiPrvSE.exe`	Winmgmt
`spoolsv.exe`	Spooler
`SearchIndexer.exe`	WSearch
`wuauserv`	Windows Update

🔄 If Process Keeps Returning - Check All Respawn Sources

Some stubborn processes have multiple sources that can restart them. Check all four locations systematically.

A Services

Windows Services are the most common source of process respawning. A service can automatically restart its associated process.

> **Find service for a process**

```
tasklist /SVC /FI "IMAGENAME eq msiexec.exe"
```

> **List all running services**

```
sc query state= all
```

> **Stop and disable a service**

```
:: Stop the service
net stop servicename

:: Disable the service permanently
sc config servicename start= disabled
```

> **GUI Method**

```
:: Open Services Manager
services.msc

:: Then: Find service → Right-click → Properties → Startup type: Disa
```

B

Scheduled Tasks

Scheduled Tasks can run processes at specific times, on startup, or triggered by events. They often restart processes periodically.

> Search for scheduled tasks containing process name

```
schtasks /query /fo LIST /v | findstr /i "msiexec"
```

> List all scheduled tasks

```
schtasks /query /fo TABLE
```

> Disable a scheduled task

```
schtasks /change /tn "TaskName" /disable
```

> Delete a scheduled task

```
schtasks /delete /tn "TaskName" /f
```

> PowerShell - Find tasks running specific process

```
Get-ScheduledTask | Where-Object {$_.Actions.Execute -like "*msiexec*
```

> GUI Method

```
:: Open Task Scheduler
taskschd.msc

:: Then: Browse Task Scheduler Library → Find task → Right-click → Di
```

C Startup Registry Entries

Programs can register themselves in the Windows Registry to start automatically when Windows boots or a user logs in.

📍 Key Registry Locations

- `HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Run` - All users
- `HKCU\SOFTWARE\Microsoft\Windows\CurrentVersion\Run` - Current user
- `HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\RunOnce` - Run once (all users)
- `HKCU\SOFTWARE\Microsoft\Windows\CurrentVersion\RunOnce` - Run once (current user)

> View startup entries (All Users)

```
reg query HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Run
```

> View startup entries (Current User)

```
reg query HKCU\SOFTWARE\Microsoft\Windows\CurrentVersion\Run
```

> Search for specific process in registry

```
reg query HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Run /s | fir
reg query HKCU\SOFTWARE\Microsoft\Windows\CurrentVersion\Run /s | fir
```

> **Delete a startup entry**

```
:: Delete from All Users startup
reg delete HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Run /v "Ent

:: Delete from Current User startup
reg delete HKCU\SOFTWARE\Microsoft\Windows\CurrentVersion\Run /v "Ent
```

> **GUI Methods**

```
:: Method 1: Task Manager
Ctrl + Shift + Esc → Startup tab → Select item → Disable

:: Method 2: MSConfig
msconfig → Startup tab → Uncheck items

:: Method 3: Registry Editor
regedit → Navigate to Run keys → Delete entry
```

D Group Policy

Group Policy can force programs to run at startup or login, especially in corporate/domain environments. These settings override local preferences.

> **Open Group Policy Editor**

```
gpedit.msc
```

📍 **Group Policy Startup Locations**

- **Computer Configuration** → Windows Settings → Scripts → Startup
- **User Configuration** → Windows Settings → Scripts → Logon
- **Computer Configuration** → Administrative Templates → System → Logon → Run these programs at user logon

> **View applied Group Policies**

```
:: Generate full GP report
gpresult /h gpreport.html

:: View in console
gpresult /r
```

> **Check GP startup scripts location**

```
:: Machine startup scripts
dir C:\Windows\System32\GroupPolicy\Machine\Scripts\Startup

:: User logon scripts
dir C:\Windows\System32\GroupPolicy\User\Scripts\Logon
```

> **Force Group Policy refresh after changes**

```
gpupdate /force
```

⚠️ Domain Environment Note

In corporate/domain environments, Group Policies are controlled by IT administrators. Local changes may be overwritten by domain policies. Contact your IT department for persistent changes.

✅ Systematic Checklist - Stop Stubborn Processes

> **Complete Respawn Source Check**

```
:: ═══════════════════════════════════════════════════════════
::  CHECK ALL RESPAWN SOURCES FOR: [ProcessName]
:: ═══════════════════════════════════════════════════════════

:: [A] SERVICES
tasklist /SVC /FI "IMAGENAME eq processname.exe"
sc query servicename
net stop servicename
sc config servicename start= disabled

:: [B] SCHEDULED TASKS
schtasks /query /fo LIST /v | findstr /i "processname"
schtasks /change /tn "TaskName" /disable

:: [C] STARTUP REGISTRY
reg query HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Run | findstr
reg query HKCU\SOFTWARE\Microsoft\Windows\CurrentVersion\Run | findstr
reg delete HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Run /v "Entr

:: [D] GROUP POLICY
gpresult /r
gpedit.msc

:: [FINAL] KILL PROCESS
taskkill /IM processname.exe /T /F

:: [VERIFY] CONFIRM IT'S GONE
tasklist /FI "IMAGENAME eq processname.exe"
```

⚠️ Important Warnings

- **msiexec.exe is critical** - Disabling it prevents software installations. Re-enable when needed.
- **Always check all four sources** if a process keeps coming back after termination.
- **For extremely stubborn processes**, boot into Safe Mode to disable services and registry entries.
- **Create a System Restore point** before making registry changes.

This guide works for any Windows process — just replace `msiexec.exe` and `msiserver` with your target process and service names.

Created with ❤️ for Windows Administrators